

A Lightweight Adaptive Multimodal 3D CNN for ADHD Classification Using Structural MRI and Phenotypic Features

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Abstract. Diagnosis of Attention-Deficit/Hyperactivity Disorder (ADHD) relies primarily on behavioral and clinical assessment, motivating neuroimaging-based computational approaches; however, many deep learning models for neuroimaging have substantial parameter and computational requirements. This work presents *ADHDNet*, a lightweight multimodal 3D convolutional neural network that combines structural magnetic resonance imaging (sMRI) with phenotypic information. ADHDNet uses depthwise-separable 3D convolutions, a single 3D efficient channel attention module, and adaptive gated fusion of MRI and phenotypic representations. On a multisite ADHD-200 cohort of 497 subjects, with volumes resized to $64 \times 64 \times 64$ and five phenotypic features, ADHDNet (71,552 parameters; 0.19 GFLOPs) achieved 90.00% accuracy and an AUC of 0.9543 on a stratified held-out validation set and $87.52 \pm 4.61\%$ accuracy (AUC 0.951 ± 0.017) in 5-fold cross-validation. Compared with a conventional Conv3D multimodal model (1.376M parameters), it uses 94.8% fewer parameters and 97.2% fewer FLOPs at comparable cross-validated AUC (0.957 ± 0.020). A component-wise ablation shows, however, that the phenotypic branch supplies most of the discriminative information, as the MRI-only variant performed close to chance.

Keywords: ADHD · Structural MRI · Multimodal learning · 3D CNN · Depthwise-separable convolution · Lightweight deep learning

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by persistent inattention, hyperactivity, and impulsivity. Its diagnosis relies primarily on behavioral and clinical evaluation, which is challenging because ADHD symptoms overlap with those of other developmental and psychiatric conditions. This has motivated neuroimaging-based computational approaches for identifying brain patterns associated with ADHD [1]–[3].

Structural magnetic resonance imaging (sMRI) provides non-invasive information about brain anatomy and has been widely used for computational ADHD analysis [4]. Earlier studies relied on manually derived measurements, such as

gray-matter volume and cortical thickness, combined with personal characteristics; Lohani and Rana [5], for example, reported a maximum accuracy of 75% with conventional classifiers. Deep learning has since enabled direct learning of imaging representations, with recent ADHD studies exploring convolutional, multimodal, attention-based, and ensemble architectures [6]–[9]. Increasing architectural complexity, however, also increases the number of trainable parameters and the computational cost, which makes parameter efficiency an important consideration for resource-constrained settings. Deepika et al. [10] explicitly addressed this with a lightweight attention-driven KAN that reached 79.25% accuracy with only a few thousand parameters.

Relatively little work has investigated parameter-efficient multimodal architectures that directly process volumetric structural MRI together with phenotypic information. This study therefore investigates whether a compact 3D architecture can jointly model structural MRI and phenotypic information while substantially reducing parameter requirements. To this end, we propose *AD-HDNet*, a lightweight adaptive multimodal 3D CNN. The main contributions are:

1. A **lightweight 3D MRI encoder** based on depthwise-separable Conv3D for parameter-efficient volumetric representation learning.
2. Integration of a single **3D efficient channel attention (ECA)** module for channel-wise recalibration of the encoder output.
3. An **adaptive gated fusion** mechanism for sample-dependent integration of MRI and phenotypic representations.
4. An **evaluation of parameter and computational efficiency** against a conventional Conv3D multimodal model on a multisite ADHD-200 cohort, complemented by 5-fold cross-validation and a component-wise ablation study.

2 Related Work

2.1 MRI-Based and Multimodal ADHD Classification

Earlier approaches to MRI-based ADHD classification generally extracted neuroanatomical or functional-connectivity features and applied conventional classifiers [1]–[3]. Lohani and Rana [5] combined gray-matter volume and cortical-thickness features with personal characteristics and achieved 75% accuracy with repeated ten-fold crossvalidation. More recent studies learn representations directly from neuroimaging data: CNNs have been applied to fMRI [6], [11], graph-theoretical features of resting-state networks have been combined with ensemble classifiers [12], graph convolutional networks with multi-head attention have been applied to hybrid-order brain networks [13], and Pedrollo et al. [9] reported CNN ensemble-based classification of resting-state fMRI. Phenotypic variables such as age, sex, medication status, and clinical ratings can complement imaging features: Deepika et al. [14] combined resting-state functional connectivity with phenotypic data, raising accuracy to 77.59%. A recent preprint has also explored

Table 1. Selected recent ADHD classification studies on ADHD-200 data. Accuracies are not directly comparable because modalities and protocols differ.

Study	Year	Modality	Model	Params	Acc. (%)	Limitations
Lohani & Rana [5]	2023	sMRI + personal char.	SVM/RF/KNN	NR	75.0	Handcrafted morphometric features; conventional classifiers
Gülhan & Özmen [11]	2024	fMRI	3D CNN	NR	76.19 [†]	Single modality; accuracy varies by site (69.9–76.2%)
Deepika et al. [14]	2024	rs-fMRI + phenotypic	Multimodality model	NR	77.59	Depends on atlas and connectivity choices; no efficiency analysis
Zamanzadeh et al. [12]	2024	rs-fMRI	EasyEnsemble	NR	74.30	Handcrafted graph features of one network; single modality
Wu et al. [13]	2025	rs-fMRI	HAGCN	NR	77.95	Requires brain-network construction; single modality
Deepika et al. [10]	2025	MRI-derived features	Lightweight attention-KAN	Few thousand	79.25	Operates on derived features rather than volumetric images
ADHDNet (ours)	2026	3D sMRI + phenotype	DS-Conv3D + ECA + gated fusion	0.0716M	90.0[‡]	Performance driven mainly by phenotypic features; no external test set

NR: not reported. [†]Best per-site accuracy (NeuroIMAGE 76.19%, NYU 69.92%, PU 70.77%).[‡]Held-out split; 5-fold cross-validation: $87.52 \pm 4.61\%$.

3D structural MRI and clinical phenotypes with adaptive gated fusion, reporting 87% accuracy on the Peking-1 subset of ADHD-200.

2.2 Lightweight ADHD Models

Model complexity matters when systems are intended for computationally constrained environments. Deepika et al. [10] explicitly targeted parameter efficiency with an attention-driven KAN using only a few thousand parameters. Following this direction, the present study treats parameter efficiency as a primary design objective. Table 1 summarizes representative recent studies and their main limitations.

2.3 Research Gap

Existing studies have shown the value of multimodal information and, separately, of parameter-efficient design, but few combine both by processing volumetric structural MRI together with phenotypic information under an explicit parameter budget. Most also do not report model size or computational cost (Table 1). ADHDNet is designed to address this combination.

3 Materials and Methods

3.1 Dataset

Experiments used the preprocessed anatomical component of the publicly available ADHD-200 dataset [15], [16], comprising 575 successfully processed subjects

(222 ADHD, 353 controls) from six sites. The Pittsburgh site contained 78 controls but no ADHD subjects; since it does not provide both classes required for binary discrimination, it was excluded. The final cohort comprised 497 subjects (222 ADHD, 275 controls) from five sites, as detailed in Table 2.

Table 2. Subject distribution of the ADHD-200 anatomical data by acquisition site. The Pittsburgh site was excluded because it contains no ADHD subjects.

Site	ADHD	Control	Total	ADHD (%)	Status
KKI	22	60	82	26.8	Included
NYU	115	96	211	54.5	Included
OHSU	36	35	71	50.7	Included
Peking_1	24	61	85	28.2	Included
NeuroIMAGE	25	23	48	52.1	Included
Pittsburgh	0	78	78	0.0	Excluded
Final cohort	222	275	497	44.7	

3.2 Preprocessing and Phenotypic Features

Each NIfTI volume was resized to $64 \times 64 \times 64$ voxels by trilinear interpolation and min-max normalized to $[0, 1]$. No additional skull stripping, registration, bias correction, or segmentation was applied, to keep the pipeline compact and reproducible. Five phenotypic variables were used: age, sex, medication status, site, and ADHD Index. Missing values (coded -999) were imputed with the training-set median (continuous) or mode (categorical); age and ADHD Index were z-scored with training-set statistics, and categorical variables were scaled to $[0, 1]$.

4 Proposed ADHDNet

ADHDNet consists of a lightweight 3D MRI encoder, a compact phenotypic encoder, and an adaptive gated fusion module (Fig. 1). The complete model contains 71,552 trainable parameters.

Lightweight 3D MRI Encoder. The encoder consists of three blocks with channel progression $1 \rightarrow 24 \rightarrow 48 \rightarrow 96$. Each block applies a depthwise-separable convolution [17], [18] (a $3 \times 3 \times 3$ depthwise kernel followed by a $1 \times 1 \times 1$ pointwise convolution), batch normalization [19], ReLU, and $2 \times 2 \times 2$ max pooling, so the resolution falls from 64^3 to 8^3 . Whereas a standard 3D convolution with kernel size K requires $K^3 C_{in} C_{out}$ weights, the depthwise-separable form requires only $K^3 C_{in} + C_{in} C_{out}$. The output feature map is recalibrated by 3D ECA, globally average-pooled to 96 dimensions, and projected to a 128-dimensional embedding z_{MRI} .

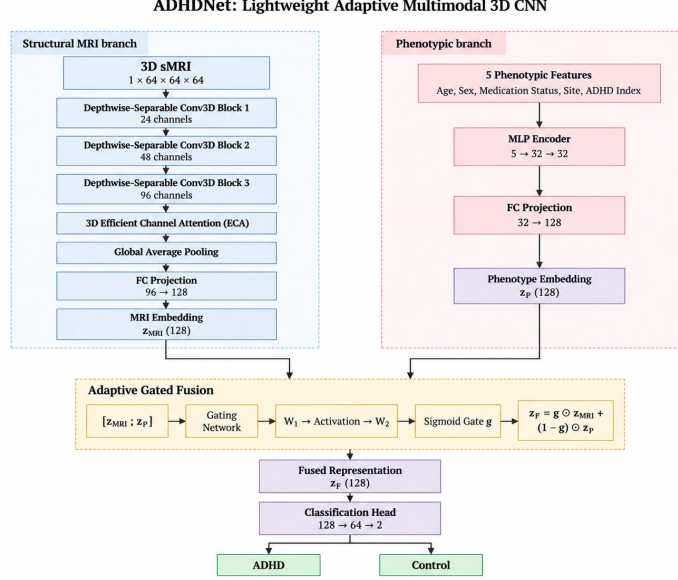


Fig. 1. Architecture of ADHDNet: a depthwise-separable 3D MRI encoder with 3D ECA, an MLP phenotypic encoder, and adaptive gated fusion followed by a classification head.

3D Efficient Channel Attention. A single 3D extension of ECA [20] is applied to the 96×8^3 output of the third block, computing $F_{att} = A(F) \odot F$. The channel weights $A(F)$ are obtained by global average pooling, a 1D convolution across channels with adaptively chosen kernel size ($k = 3$ for 96 channels), and a sigmoid, so the module adds only three parameters.

Phenotypic Encoder and Adaptive Gated Fusion. In parallel, a $5 \rightarrow 32 \rightarrow 32$ MLP with batch normalization, ReLU, and dropout [21] encodes the phenotypic features into $h_P \in \mathbb{R}^{32}$. In the spirit of gated multimodal units [22], a gating network computes $g = \sigma(W_2 \delta(W_1[z_{MRI}; h_P])) \in \mathbb{R}^{128}$ from the 160-dimensional concatenation, where σ is the sigmoid and δ the ReLU. With $z_P \in \mathbb{R}^{128}$ a linear projection of h_P , the fused representation is

$$z_F = g \odot z_{MRI} + (1 - g) \odot z_P, \quad (1)$$

so the relative contribution of each modality is determined per sample. A $128 \rightarrow 64 \rightarrow 2$ classification head with dropout produces the class probabilities.

5 Experimental Setup

Training. Models were implemented in PyTorch [23] and trained with AdamW [24] (learning rate 10^{-4} , weight decay 10^{-4}), batch size 8, at most

Table 3. Held-out validation performance of ADHDNet ($n = 100$; ADHD positive).

Accuracy	AUC	Sensitivity	Specificity	Precision	NPV	F1-score
90.00%	0.9543	95.56%	85.45%	84.31%	95.92%	0.8958

150 epochs with early stopping (patience 15), class-weighted cross-entropy with label smoothing [25] of 0.1, gradient clipping at 1.0, and cosine annealing [26]. Training volumes were randomly flipped along each axis with probability 0.5, the checkpoint with the highest validation AUC was retained, and all random seeds were fixed to 42.

Comparator. A conventional Conv3D multimodal model was implemented in the same framework: four standard Conv3D blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels) with 3D ECA after blocks 2–4, generalized-mean pooling, a 256-dimensional embedding, a $5 \rightarrow 32 \rightarrow 64 \rightarrow 64$ phenotypic MLP, the same gated fusion, and a $256 \rightarrow 128 \rightarrow 64 \rightarrow 2$ head (1,376,080 parameters).

Evaluation Protocol. (i) *Held-out validation:* a stratified 80:20 split yielded 397 training and 100 validation subjects (45 ADHD, 55 controls). Because the validation set was also used for early stopping and model selection, these results are held-out validation, not independent test, performance. (ii) *Cross-validation:* ADHDNet and the comparator were evaluated with stratified 5-fold cross-validation on all 497 subjects (at most 60 epochs, patience 10), taking fold metrics at the epoch of best validation AUC. (iii) *Ablation:* variants were trained on the same held-out split with the same budget. In all settings, imputation and normalization statistics were fitted on training subjects only. Accuracy, AUC, sensitivity, specificity, precision, negative predictive value (NPV), and F1-score were computed with ADHD as the positive class. FLOPs were counted with PyTorch’s `FlopCounterMode` for one 64^3 volume, and latency was averaged over 50 runs after 10 warm-up passes on a CUDA GPU.

6 Results

6.1 Held-Out Classification Performance

On the held-out validation set (Table 3), ADHDNet achieved 90.00% accuracy and an AUC of 0.9543 (best epoch 60; early stopping at epoch 75). It correctly classified 43 of 45 ADHD subjects and 47 of 55 controls (Fig. 2). Of the 10 errors, 8 were false positives and 2 false negatives, so sensitivity (95.56%) exceeded specificity (85.45%) and NPV (95.92%) exceeded precision (84.31%): on this cohort, a negative prediction was more reliable than a positive one.



Fig. 2. Held-out validation results ($n = 100$): (a) ROC curve (AUC = 0.9543); (b) confusion matrix at threshold 0.5.

Table 4. ADHDNet versus the conventional Conv3D comparator: model complexity per 64^3 volume and 5-fold cross-validation (mean $\pm$ SD). FP32 weights = parameters $\times$ 4 bytes; latency at batch size 1, throughput at batch size 8.

	Measure	Conv3D comparator	ADHDNet
Complexity	Parameters	1,376,080	71,552
	FP32 weights	5.25 MiB	0.27 MiB
	GFLOPs	6.7952	0.1932
	Latency (ms)	4.32 ± 0.12	2.36 ± 0.15
	Throughput (vol/s)	242.6	662.9
5-fold CV	Accuracy (%)	86.51 ± 3.64	87.52 ± 4.61
	AUC	0.9566 ± 0.0197	0.9512 ± 0.0169
	Sensitivity (%)	85.10 ± 9.63	87.36 ± 5.75
	Specificity (%)	87.64 ± 7.09	87.64 ± 4.71
	F1-score (%)	84.79 ± 4.70	86.21 ± 5.20

6.2 Cross-Validation and Computational Efficiency

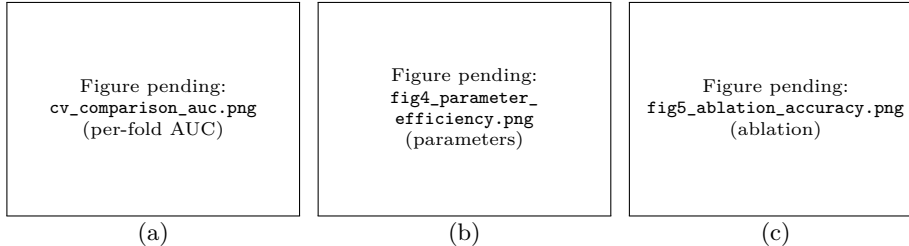
Table 4 compares ADHDNet with the conventional Conv3D comparator. In 5-fold cross-validation, ADHDNet achieved $87.52 \pm 4.61\%$ accuracy (per-fold range 79.80–90.91%) and an AUC of 0.9512 ± 0.0169 , against $86.51 \pm 3.64\%$ and 0.9566 ± 0.0197 for the comparator (Fig. 3a). Discrimination was therefore comparable, while ADHDNet had a slightly higher mean accuracy, sensitivity, and F1-score and a less variable sensitivity. The held-out accuracy of 90.00% lies at the upper end of the cross-validated range, so 87.52% is the more representative estimate. ADHDNet uses 94.8% fewer parameters ($19.2\times$) and 97.2% fewer FLOPs ($35.2\times$), reducing FP32 weight storage from 5.25 to 0.27 MiB (Fig. 3b). Latency fell by $1.8\times$ and batched throughput rose by $2.7\times$; the speed-up is smaller than the FLOP reduction because small 3D models do not saturate the GPU.

6.3 Ablation Study

Four variants each change one component of the full model (A0): removing ECA (A1); removing the phenotypic branch so that the MRI embedding feeds the

Table 5. Component-wise ablation on the held-out validation set ($n = 100$).

Variant	ECA	Pheno.	Fusion	Params	Acc. (%)	AUC	F1
A0 Full ADHDNet	✓	✓	Gated	71,552	90.00	0.9543	0.8958
A1 w/o ECA	–	✓	Gated	71,549	87.00	0.9487	0.8632
A2 w/o phenotype	✓	–	–	28,896	54.00	0.5644	0.4773
A3 Concat. fusion	✓	✓	Concat.	66,272	89.00	0.9564	0.8842
A4 Standard Conv3D	✓	✓	Gated	220,133	89.00	0.9547	0.8842

**Fig. 3.** (a) Per-fold AUC in 5-fold cross-validation; (b) trainable parameters of ADHDNet and the Conv3D comparator; (c) held-out accuracy of the ablation variants.

classifier directly (A2); replacing gated fusion with concatenation and a linear projection to 128 dimensions (A3); and replacing depthwise-separable with standard convolutions (A4). Results are shown in Table 5 and Fig. 3c.

The phenotypic branch had by far the largest effect: without it (A2), accuracy fell to 54.00% and AUC to 0.5644, close to chance, indicating that under the present preprocessing the MRI branch alone provides little discriminative information. Removing ECA (A1) reduced accuracy by 3.0 points and AUC by 0.0056 for a module with three parameters. Concatenation (A3) gave slightly lower accuracy but slightly higher AUC than gating, so this split shows no clear advantage of gated fusion. Standard convolutions (A4) matched the full model with $3.1\times$ more parameters, confirming that depthwise-separable factorization reduces model size without loss of accuracy. With 100 validation subjects, one subject corresponds to one percentage point, so differences of a few points should be interpreted with caution.

7 Discussion

ADHDNet jointly models structural MRI and phenotypic information at a small fraction of the cost of a conventional 3D model, reaching comparable cross-validated discrimination with 94.8% fewer parameters and 97.2% fewer FLOPs. The ablation attributes this efficiency mainly to depthwise-separable convolution, while the three-parameter ECA module yielded a small accuracy gain and gated fusion performed on par with concatenation.

The most important finding, however, is the dominance of the phenotypic branch: the MRI-only variant performed close to chance, whereas every variant retaining phenotypic input achieved an AUC above 0.94. The phenotypic input includes the ADHD Index, a clinical symptom rating closely related to the diagnostic label, and site, which is correlated with class proportions (Table 2). The reported accuracy should therefore not be read as evidence that structural MRI alone, as represented here, discriminates ADHD, nor that ADHDNet outperforms prior studies, which use different modalities and protocols (Table 1). The contribution is instead a *parameter-efficient multimodal 3D architecture* together with an evaluation that makes the contribution of each component explicit.

Limitations. Performance depends predominantly on phenotypic features, potentially the ADHD Index; held-out, cross-validation, and ablation metrics were taken at the epoch of best validation AUC and no independent external test set was used; cross-site generalization was not evaluated; the ablation used a single split without repeated runs; the simple preprocessing may limit the anatomical information available to the MRI branch; and latency was measured on a single GPU.

8 Conclusion

This paper presented ADHDNet, a lightweight multimodal 3D CNN that combines depthwise-separable convolution, efficient channel attention, and adaptive gated fusion of structural MRI and phenotypic information. On 497 ADHD-200 subjects, it achieved 90.00% held-out accuracy (AUC 0.9543) and $87.52 \pm 4.61\%$ cross-validated accuracy with 71,552 parameters and 0.19 GFLOPs, 94.8% fewer parameters and 97.2% fewer FLOPs than a conventional Conv3D comparator at comparable AUC. The ablation showed that the phenotypic branch supplies most of the discriminative information. Future work will evaluate the model without the ADHD Index and site, strengthen the MRI representation through improved preprocessing, and perform cross-site and external validation.

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